Battery literature digest — week of 09 November 2026

8 papers selected from 1081 scanned. Scores out of 10; click a title to open the paper.

1. Operando Attenuated Total Reflectance Far-Ultraviolet Spectroscopic Analysis at the Cathode/Electrolyte Interface

8/10 · *Angewandte Chemie International Edition* · operando ATR-FUV spectroscopy with PCA and quantum chemical calculations

This is a new operando far-UV ATR method for the buried cathode/electrolyte interface. PCA isolates a voltage-dependent component that appears at about 3.9 V, well below the electrolyte oxidation onset. Calculations assign it to dehydrogenated EC and VC, which points to chemical proton abstraction by reactive surface oxygen after Li+ deintercalation. The assignment depends on calculations, so it is less direct than the method itself.

2. In Situ Nanoscale Visualization of Stripping-Induced Failure Behavior in Aqueous Zinc Batteries

8/10 · *ACS Nano* · in situ nanoscale transmission X-ray microscopy (4th-gen synchrotron) with optical flow analysis

Operando nano-X-ray microscopy follows Zn plating/stripping, including the planar-to-mossy transition at 50 mA cm-2. Optical flow analysis of stripping classifies residual (“dead”) Zn into four origins: plating dendrites, crystallographically selective stripping, electrical isolation, and inward-dissolution fragments. Ex situ XRD and DFT support preferential dissolution of Zn(100) and (101) over (002).

3. Operando Quantifying Li Plating Processes in Lithium-Ion Pouch Cells

7/10 · *Angewandte Chemie International Edition* · operando dynamic EIS with operando solid-state NMR, mass spectrometry titration

Dynamic EIS defines a critical Li-plating areal capacity, validated against operando ssNMR and MS titration in pouch cells (R^2>0.97, 0–45 °C). A transmission line model links it to ~10% graphite surface coverage by Li nuclei, the switch from nucleation to growth control, and a temperature-driven change in nucleation mode (3DP to 2DI). The coverage figure is model-derived, not imaged.

4. Hydrogen-Bonding Order Associated With Radical Delocalization in Layered Organic Cathode Materials

6/10 · *Angewandte Chemie International Edition* · MicroED, solid-state 15N/13C NMR, synchrotron x-ray scattering, pulsed EPR

Resolves a structural misassignment: “TAQ” is the same pyrazine-based molecule as TAPT, not piperazine, and the conductivity gap of two orders of magnitude comes from crystallinity and ordered hydrogen bonding. EPR links this order to delocalized π-spin. It is a strong multi-technique structural study, but it is on pristine materials, not operando or imaging.

5. Facet-dependent molecular adsorption modulates zinc anode work function for the favourable orientation of zinc electrocrystallization

6/10 · *Nature Chemistry* · work function measurements (likely KPFM/UPS), texture analysis across 29 additives

Proposes facet-dependent work function change on additive adsorption, via Fermi-level equilibration, as a descriptor of Zn deposit texture. Across 29 additives it predicts orientation with 93.1% accuracy. This is a rare mechanistic, predictive framework for additive screening, though it is not imaging-based and the abstract gives few details.

6. Gas Retention in Porous Cathodes Governs Bicarbonate Electrolysis Performance

6/10 · *ACS Energy Letters* · operando X-ray imaging (radiography) of gas in porous cathode

Operando X-ray imaging directly visualizes gas retention in liquid-fed carbon paper/cloth cathodes during bicarbonate electrolysis, identifying it as a reversible transport limitation. Flow-pulse recovery tests link it to performance, and an interdigitated flow field raised formate partial current density to 304 mA cm–2. This is electrolyzer rather than battery work, but the imaging approach transfers.

7. Direct Evidence of Intermediate Oxygen Species Evolution in Cathode Oxides With Nominally Stable Superstructure

6/10 · *Angewandte Chemie International Edition* · Likely RIXS / spectroscopy of oxygen redox (ex situ), ribbon-ordered P2/P3 Na cathodes

Tracks oxidised oxygen speciation over cycling in ribbon-ordered P2/P3 $Na_{0.6}Li_{0.2}Mn_{0.8}O_2$. It reports a shift from O_2^- holes and $(O_2)_n^-$ dimers to trapped molecular O_2, coupled with Li-vacancy cluster growth and Mn-rich domain formation. It also shows the same trend in P3, where Mn migration is restricted. Spectroscopic and mostly ex situ, but a clear mechanistic claim.

8. Quantifying the Dissolution of Organic and Inorganic Components in Solid Electrolyte Interphase for Sodium-Ion Batteries

6/10 · *Angewandte Chemie International Edition* · ion chromatography + EQCM + XPS (component-resolved SEI dissolution)

Quantifies absolute retention of organic and inorganic SEI components on hard carbon, not just aggregate loss. Tetraglyme cuts total SEI dissolution from 75% to 53% and raises retained F-species from 18% to 30%. The anion-rich solvation shell is the proposed cause. The method is useful, but it is not spatially resolved and the electrolyte tuning is partly incremental.